

Power Efficiency: $\eta = \frac{P_{out}}{P_{in}} = \frac{P_{loss1}}{P_{loss0}} = \frac{P_{in} - P_{out1}}{P_{in} - P_{out2}} = \frac{1 - D_1}{1 - D_0}$

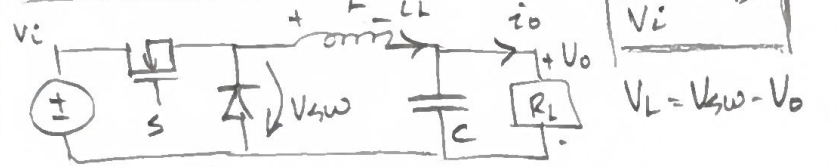
Linear Mode Converter:

$V_i \rightarrow R \rightarrow V_o$ $\frac{V_o}{V_i} = \frac{R_{out}}{R + R_{out}}$, $\eta = \frac{R_{out}}{R + R_{out}}$

• efficiency is high only if V_{out} is close to V_{in}

• poor regulation since V_{out} changes if R_{out} changes

Buck Converter



$\frac{V_o}{V_i} = D$

$V_L = V_{SW} - V_o$

L-volt-sec balance:

$[i_L]_{t=0} = [i_L]_{t=T_{SW}}$

$S_1: [i_L]_0^{DT_{SW}} = \left[\frac{V_{in} - V_{out}}{L} \right]_0^{DT_{SW}}$

$S_0: [i_L]_{DT_{SW}}^{T_{SW}} = \left[-\frac{V_{out}}{L} \right]_{DT_{SW}}^{T_{SW}}$

max when $D = 0.5$

$\Delta i_{Lppk} = [i_L]_{DT_{SW}} - [i_L]_0 = (1-D)D \frac{V_{in}}{L} T_{SW}$

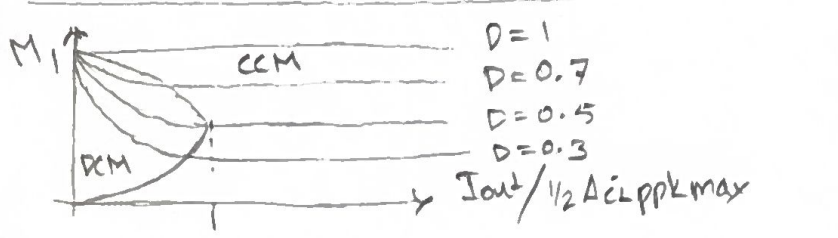
CCM: $\frac{1}{2} \Delta i_{Lppk} < I_{out}$

DCM: $\frac{1}{2} \Delta i_{Lppk} > I_{out}$

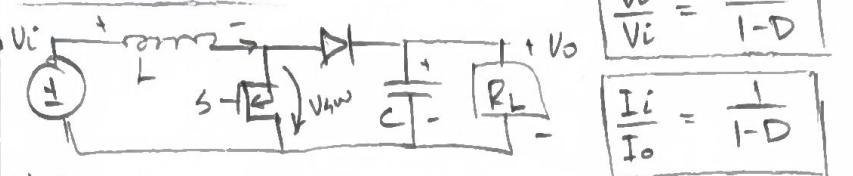
critical load resistance: $R_{out} > \frac{2L}{(1-D)T_{SW}}$

$\Delta i_{Lppk} = \left(\frac{V_{in} - V_{out}}{L} \right) DT_{SW}$

$M = \frac{2}{1 + \sqrt{1 + \frac{4K}{D^2}}}$, $K = \frac{2L}{T_{SW} \cdot R_{out}}$



Boost Converter



$\frac{V_o}{V_i} = \frac{1}{1-D}$

$\frac{I_i}{I_o} = \frac{1}{1-D}$

L-volt-sec balance:

$S_1: [i_L]_0^{DT_{SW}} = \left[\frac{V_{in}}{L} \right]_0^{DT_{SW}}$

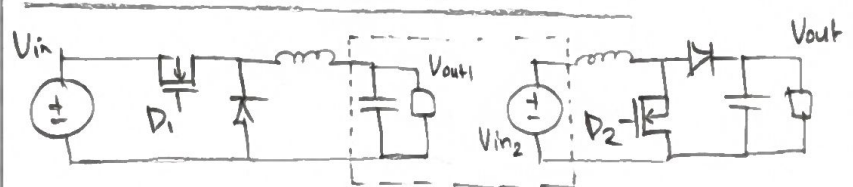
$S_0: [i_L]_{DT_{SW}}^{T_{SW}} = \left[\frac{V_{in} - V_{out}}{L} \right]_{DT_{SW}}^{T_{SW}}$

C-q. balance:

$(I_{out})DT_{SW} = (I_{in} - I_{out})(1-D)T_{SW}$

$I_{out} < I_{in}$

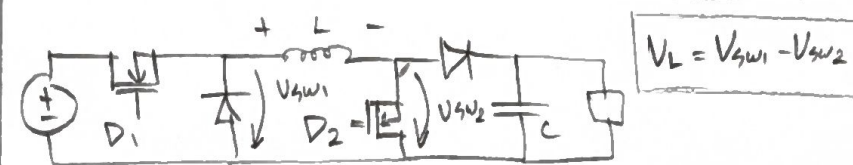
4-switch Buck Boost Converter



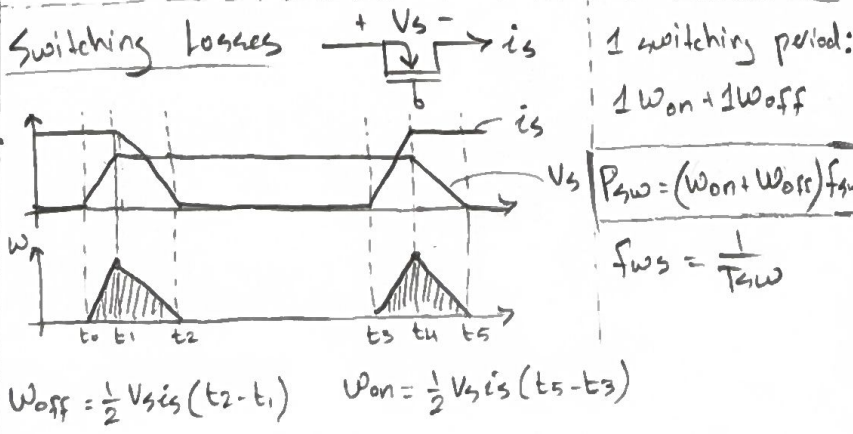
From Buck: $\frac{V_{out1}}{V_{in}} = D_1 \Rightarrow V_{out1} = D_1 V_{in}$

From cascade: $V_{in2} = V_{out1}$

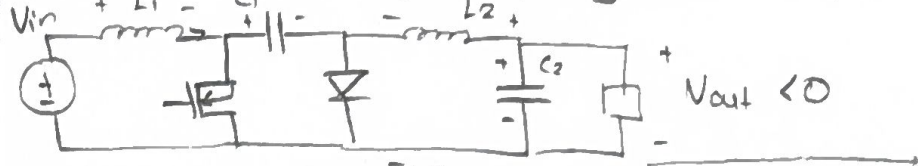
From Boost: $\frac{V_{out}}{V_{in2}} = \frac{1}{1-D_2} \Rightarrow \frac{V_{out}}{V_{in}} = \frac{D_1}{1-D_2}$



can vary V_{SW1} , V_{SW2} thru D_1 , D_2 to get $V_L \Rightarrow i_L$



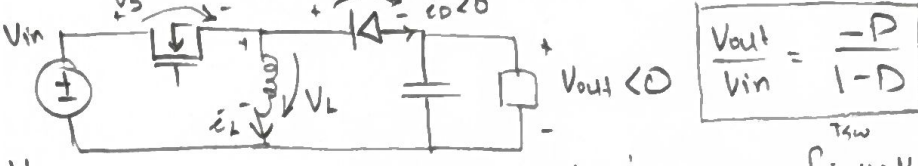
Cuk Converter: only use 2 pwr semi's, but penalty is negative polarity output voltage



$\langle V_L \rangle = 0 \Rightarrow \int_0^{T_{SW}} V_L(t) dt = 0$

$\frac{V_{out}}{V_{in}} = \frac{-D}{1-D}$

Inverting Buck-Boost: further reduce magnet. comp. but each higher rating



$\frac{V_{out}}{V_{in}} = \frac{-D}{1-D}$

$\langle i_C \rangle = 0 = \int_0^{T_{SW}} i_C(t) dt$

find $\langle i_L \rangle = ?$

$S_1: i_C = -i_{out}$

$S_0: i_C = -i_L - i_{out}$

$i_L = \left(\frac{1}{1-D} \right) i_{out}$

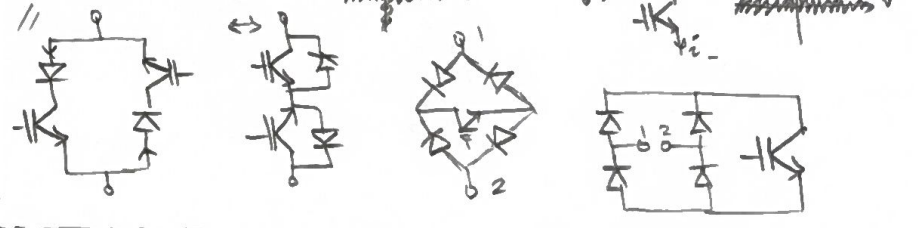
$\langle i_L \rangle = -\left(\frac{1}{1-D} \right) I_{out}$

> 0

Diode: I_{GBT} I/V bidirectional Δw



4-Quadrant Sw:



Conducⁿ Losses:

$P_{cond} = I_{rms}^2 \cdot R_{on}$

$I_{rms} = \sqrt{\frac{1}{T_{SW}} \int_0^{T_{SW}} i^2(t) dt}$

higher inductor ripple means higher conducⁿ losses

Basics: $i_C = C \frac{dv_C}{dt}$, $V_L = L \frac{di_L}{dt}$

$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$

$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$

$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$

$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

$\sin 2\theta = 2 \sin \theta \cos \theta$

$2 \cos^2 \theta - 1 = \cos 2\theta = 1 - 2 \sin^2 \theta$

$$v(t) = V_0 + \sum_{n=1}^{\infty} V_n \cos(n\omega t + \phi_n)$$

$$i(t) = I_0 + \sum_{m=1}^{\infty} I_m \cos(m\omega t + \theta_m)$$

$$V_n = \sqrt{a_n^2 + b_n^2} \leftarrow a_n = \frac{2}{T} \int_0^T v(t) \cos(n\omega t) dt$$

$$b_n = \frac{2}{T} \int_0^T v(t) \sin(n\omega t) dt$$

instantaneous power: $p(t) = i(t)v(t)$

Average Power: $P_{av} = \frac{W_{total}}{T} = \frac{1}{T} \int_0^T i(t)v(t) dt$ ← 3 types of prod's

① DC x DC: $V_0 I_0$

② DC x sinusoid: 0

③ sinusoid x sinusoid:

only voltages and currents of same freq. contributes to the average pwr

0 if $m \neq n$

$\frac{V_n I_n \cos(\phi_n - \theta_m)}{2}$ if $n = m$

RMS in t:

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt}$$

$$v(t) = V_{pk} \sin(\omega t) = \sqrt{2} V_{rms} \sin(\omega t)$$

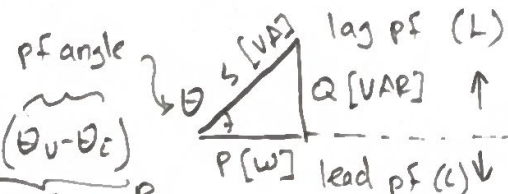
RMS in freq:

$$I_{rms} = \sqrt{I_0^2 + \sum_{n=1}^{\infty} \frac{I_n^2}{2}}$$

$$\Rightarrow V_{pk} = V_{rms} \sqrt{2}$$

$$P_{loss} = I_{rms}^2 R = \frac{V_{rms}^2}{R}$$

$$S = \bar{V}_{rms} \bar{I}_{rms}$$



$$P_{avg} = \bar{V}_{rms} \bar{I}_{rms} \cos(\theta_v - \theta_i)$$

real pwr [W] apparent pwr [VA] $PF = \frac{P}{S}$

$$PF = \frac{P_{avg}}{\bar{V}_{rms} \bar{I}_{rms}}, \quad THD = \sqrt{\sum_{n=2}^{\infty} \frac{I_n^2}{I_1^2}}, \quad \text{distortion factor} = \frac{1}{\sqrt{1 + THD^2}}$$

true pf = (distortion factor) (displacement factor)

$$= \left(\frac{1}{\sqrt{1 + THD^2}} \right) (\cos(\phi_1 - \theta_1))$$

$$V(t) = V_{rms} \sqrt{2} \cos(\omega t + \phi) \Rightarrow \bar{V} = V_{rms} \angle \phi$$

$$= V_{pk} \cos(\omega t + \phi)$$

$$V_{rms} = \sqrt{V_0^2 + \sum_{n=1}^{\infty} \frac{V_{pk,n}^2}{2}}$$

$$V_{rms} = \sqrt{\sum_{n=1}^{\infty} (V_{rms,n})^2}$$

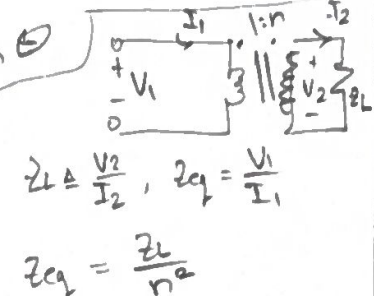
$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$\sin A \sin B = -\frac{1}{2} [\cos(A+B) - \cos(A-B)]$$

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\frac{1}{T} \int_0^T i_c(t) dt = 0$$

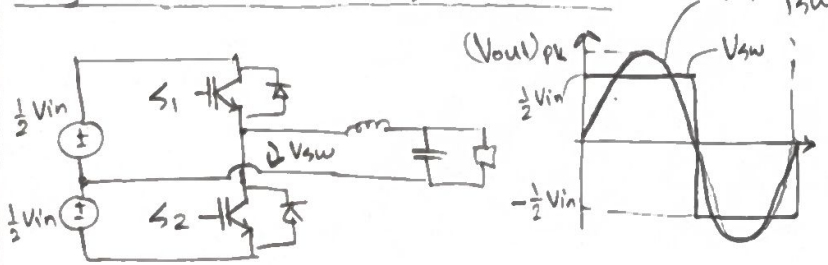
$$\frac{1}{T} \int_0^T V_L(t) dt = 0$$



$$x(t) = \sum_{n=1}^{\infty} \frac{4}{n\pi} V_{pk} \sin(n\omega t)$$

for only n odd: 1, 3, 5...

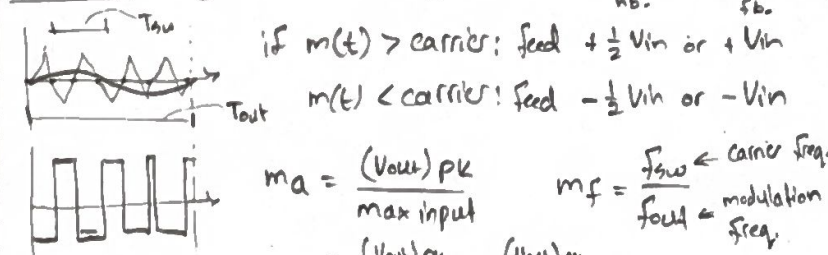
Single Phase Half-Bridge Inverter



$$V_{out} = (V_{out})_{pk} \sin(2\pi f_{out} \cdot t) = \frac{4}{\pi} \cdot \frac{1}{2} V_{in} \sin(2\pi f_{out} \cdot t)$$

S_1, S_2 can be off simultaneously but not both on.

SPWM: triangular carrier, rectangular V_{sw} , desired sinusoid $m(t)$



if $m(t) > \text{carrier}$: feed $+\frac{1}{2} V_{in}$ or $+V_{in}$

if $m(t) < \text{carrier}$: feed $-\frac{1}{2} V_{in}$ or $-V_{in}$

$$m_a = \frac{(V_{out})_{pk}}{\text{max input}} = \frac{(V_{out})_{pk}}{1/2 V_{in}} \text{ or } \frac{(V_{out})_{pk}}{V_{in}}$$

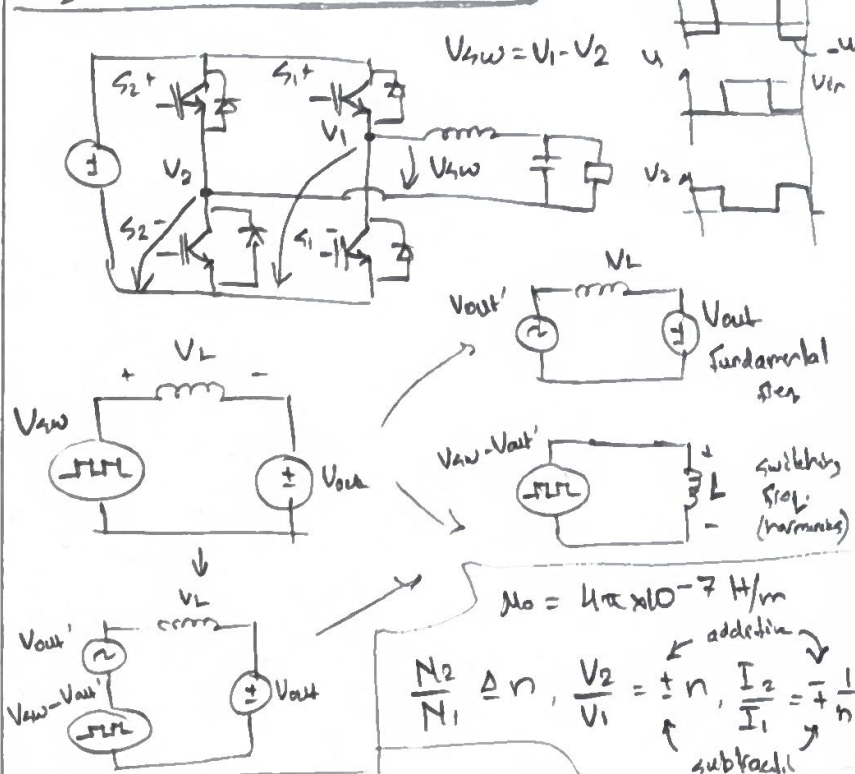
$$m_f = \frac{f_{sw}}{f_{out}} \leftarrow \begin{matrix} \text{carrier freq.} \\ \text{modulation freq.} \end{matrix}$$

harmonics order: $j m_f \pm k$

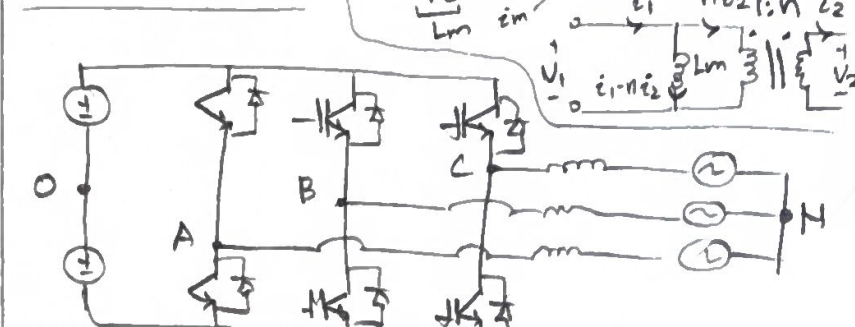
odd $j \rightarrow$ even $k \leftarrow$ preferred and happens when m_f is odd

even $j \rightarrow$ odd k

Single Phase Full Bridge Inverter



Three Phase Inverter



$$V_1 = \frac{N_1}{L_m} \frac{d}{dt} (\int i_1 - n \int i_2)$$

$$V_{no} = \frac{V_{ao} + V_{bo} + V_{co}}{3}$$

V_{no} does not drive phase (common mode voltage) current

differential mode voltage: $V_{ao} - V_{no} = V_{La} + V_{LaA}$

$\rightarrow$ drives inductor current

$$F = Ni, \quad B = \mu H = \mu_i \mu_r H, \quad F = \frac{1}{2} (R_c + R_g)$$

$$R_c = \frac{l_c}{\mu_0 \mu_r A_c}, \quad R_g = \frac{g}{\mu_0 \mu_r A_g}, \quad L = \frac{\gamma}{i} = \frac{\mu \Phi}{i}$$

$$L = \frac{N^2}{R_c + R_g}, \quad \text{if } \mu_r \rightarrow \infty, \quad L = \frac{N^2}{R_g}$$

Inductor $\downarrow$ $W = \frac{1}{2} B H$ (Volume)

$W = \frac{1}{2} \frac{B^2}{\mu_0 \mu_r} + \frac{1}{2} \frac{B^2}{\mu_0}$